Maverick Oh's CV

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PERSONAL DETAILS

Legal Name	Sang-Hyun Oh
Preferred Name	Maverick Oh
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ACADEMIC APPOINTMENT

Associate Professor — Starting Aug. 2026
Montgomery College — MD, USA

Part-Time Instructor — Feb. - May. 2025
Montgomery College

EDUCATION

PhD in Physics — Aug. 2020 - Present
University of California, Merced (UC Merced) — CA, USA — Degree expected May 2026
Advisor: Anna Nierenberg
GPA: 3.95/4.00

M. S. in Physics — Aug. 2020 - Dec. 2023
University of California, Merced (UC Merced)
Advisor: Anna Nierenberg
Awarded as a part of Ph.D. studies.

Student Exchange Program — Aug. - Dec. 2019
California Institute of Technology (Caltech) — CA, USA

Student Exchange Program — Jun. - Aug. 2016
University of California, Berkeley (UC Berkeley) — CA, USA

B. S. in Physics — Mar. 2015 - Aug. 2020
Gwangju Institute of Science and Technology (GIST) — South Korea
Advisor: Keun-Young Kim
Cum Laude (GPA: 3.97/4.5)

RESEARCH EXPERIENCE

Dark Matter Analysis with Quadruply Lensed Quasars — Doctoral Thesis Project

Supervisor | Prof. Anna Nierenberg, UC Merced

About | Developed Python code that models galaxy mass and light profiles that deviate from perfect elliptical symmetry; built and optimized semi-analytic joint population prior of the non-ellipticity parameters using PyTorch for it to be used for more robust inference. Currently working on such non-elliptical shape measurement of galaxies in COSMOS field using HST and JWST data using Python–more specifically, JAX for fast inference. Previously worked on modeling and analysis of gravitationally lensed quasar systems with lensing galaxy profiles that deviate from elliptical symmetry to understand the impact of the galaxy shape modeling in lensing. Statistically constrained the large-scale structure of galaxy mass model and its impact on small-scale structure inference, which is related to the fundamental properties of dark matter.

Publication | [1],[4, 5] **Talk** | [10, 11, 12]

Graphical Notation of Tensor Calculus — Bachelor Thesis Project

Quantum Field and Gravity Theory Group

Supervisor | Prof. Keun-Young Kim, GIST (Gwangju, South Korea)

About | Conducted research on the graphical notation of tensorial identities and equations (Penrose graphical notation, etc.), and applied it to vector calculus identities. Wrote a pedagogical paper introducing the graphical notation in vector calculus and its application on undergraduate physics.

Publication | [3],[7] **Talk** | [13, 14] **On Media** | [16]

Machine Learning on AdS/CFT

Quantum Field and Gravity Theory Group

Supervisor | Prof. Keun-Young Kim, GIST (Gwangju, South Korea)

About | Worked on deep-learning application on AdS/CFT correspondence and applied it to mechanical systems using neural network and neural ODE. Taught and guided two undergraduate student researchers.

Publication | [2]

Astrophysics Simulation Result Analysis

Theoretical Astrophysics Including Relativity and Cosmology Group (TAPIR)

Supervisors | Prof. Philip Fajardo-Hopkins and Dr. Coral Wheeler, Caltech

About | Analyzed galaxy formation history from GIZMO Feedback In Realistic Environments (FIRE) simulations with different reionization periods.

X-ray Diffraction Simulation and Data Analysis

X-ray Laboratory for Nanoscale Phenomena

Supervisor | Prof. Do Young Noh, GIST (Gwangju, South Korea)

About | Simulated X-ray diffraction (XRD) patterns with diffraction gratings and analyzed XRD data for single-pulse emission spectroscopy using MATLAB. Tasks included data preprocessing, curve fitting, and numerical studies on spectroscopy correlations.

Publication | [6]

Poetics Research on Early Korean Science Poems by Yi-Sang

Supervisor | Prof. Soo Jong Lee, GIST (Gwangju, South Korea)

About | Deciphered the complex poems of Yi-Sang, written in the 1930s, which incorporate themes from the contemporary physics of the time, such as Einstein's theory of relativity. My work involved unraveling the scientific concepts embedded in these enigmatic literary works.

Publication | [8, 9] **Talk** | [15] **On Media** | [17, 18]

PUBLICATION

First & Co-First Author

[1] **Oh, Maverick S.H.**, Anna Nierenberg, Daniel Gilman, and Simon Birrer. Joint semi-analytic multipole priors from galaxy isophotes and constraints from lensed arcs. *Journal of Cosmology and Astroparticle Physics*, 2026(03):039, mar 2026.

[2] Mugeon Song, **Maverick S. H. Oh**, Yongjun Ahn, and Keun-Young Kim. AdS/deep-learning made easy: simple examples. *Chinese Physics C*, 45(7):073111, jul 2021.

Significant Contribution

[3] Joon-Hwi Kim, **Maverick S. H. Oh**, and Keun-Young Kim. Boosting vector calculus with the graphical notation. *American journal of physics*, 89(2):200–209, 2021.

Collaborative Contribution

[4] Daniel Gilman, Simon Birrer, Anna Nierenberg, and **Maverick S. H. Oh**. Turbocharging constraints on dark matter substructure through a synthesis of strong lensing flux ratios and extended lensed arcs. *Monthly Notices of the Royal Astronomical Society*, 533(2):1687–1713, 07 2024.

[5] Ryan E Keeley, Anna M Nierenberg, Daniel Gilman, Charles Gannon, Simon Birrer, Tommaso Treu, Andrew J Benson, Xiaolong Du, K. N. Abazajian, T. Anguita, V. N. Bennert, S. G. Djorgovski, K. K. Gupta, S. F. Hoenig, A. Kusenko, C. Lemon, M. Malkan, V. Motta, L. A. Moustakas, **Maverick S. H. Oh**, D. Sluse, D. Stern, and R. H. Wechsler. Jwst lensed quasar dark matter survey–ii. strongest gravitational lensing limit on the dark matter free streaming length to date. *Monthly Notices of the Royal Astronomical Society*, 535(2):1652–1671, 2024.

[6] Muhammad Ijaz Anwar, Sung Soo Ha, Byung-Jun Hwang, Seonghyun Han, **Maverick S. H. Oh**, Mohd Faiyaz, Do Young Noh, Hyon Chol Kang, and Sunam Kim. Hard x-ray von hamos spectrometer for single-pulse emission spectroscopy. *Journal of the Korean Physical Society*, 75(7):494–497, Oct 2019. `doi.org/10.3938/jkps.75.494`.

Degree Thesis

[7] **Maverick S. H. Oh**. Graphical notation for vector calculus and its generalization. Bachelor Thesis, 2020. `library.gist.ac.kr/storage/thesis/GIST_20155110_Sang-Hyun%20Oh_20201013103856901.pdf`.

Interdisciplinary Publications

[8] **Maverick S. H. Oh** and Soo Jong Lee. Design and construction in four-dimensional space-time in yi-sang's poems :the connection between three-dimensional angle blueprint and building infinite-hexahedral-angle bodies, and dimension expansion (이상 시의 4차원 시공간 설계 및 건축 : 삼차각설계도와 건축무한육면각체의 연결, 그리고 차원 확장). *Journal of Korean Culture (JKC)*, 54:107–156, 2021. Written in Korean.

[9] **Maverick S. H. Oh** and Soo Jong Lee. Periodic boundary condition of yi-sang's poems 1 –「building infinite-hexahedral-angle bodies - diagnosis 0:1」decryption (이상 시의 주기경계조건 1–「건축무한육면각체 - 진단 0:1」의 파해). *YiSang Review (이상리뷰)*, (18):1–32, 2022. Written in Korean.

TALKS

[10] Enhancing jwst detection of low-mass halos in quadruply lensed quasars with hst imaging of extended arcs (etalk). International Astronomy Union General Assembly (IAUGA), Busan, South Korea, 2022.

[11] Quads+arc images can tell you about dark matter models. UC Merced Cosmo-Gal Astrophysics Research Symposium (online), 2022.

[12] Non-elliptical perturbations of elliptical galaxies and gravitational lensing (poster). Sierra Conference, California, United States, 2025.

[13] Vector identity proofs for everyone with diagrams. The 3rd P&P Holics Math & Physics Seminar at GIST, Gwangju, South Korea, 2018.

[14] Graphical notation seminar. Independently arranged seminar at GIST, Gwangju, South Korea, 2020.

[15] Introduction to yi-sang's poems for physicists. Invited talk for a special session of Asia Pacific Center for Theoretical Physics (APCTP) Topical Research Program (online), 2021.

TEACHING EXPERIENCE

As An Instructor

Physical Science I Feb. - May 2025
Montgomery College
Instructor of Record for an introductory Physical Science course, covering fundamental concepts in physics and astronomy. Developed and delivered lectures, created discussion and assessment materials, and coordinated hands-on laboratory activities.
Course coordinator: Prof. Teresa Peachey. Supervisors: Prof. Kedar Hardikar and Prof. Alla Webb.

Machine Learning and Physics – Undergrad Research Program Jun. - Jul. 2020
Gwangju Institute of Science and Technology
Summer program instructor for "Machine Learning and Physics" research program designed for undergraduate students through GIST Summer Undergraduate Research Fellowship (G-SURF), under the supervision of Prof. Keun-Young Kim. Planned and designed the overall program, taught undergraduate students the basics of machine learning, software skills (PyTorch) and their application on AdS/CFT problems.

Physics Using Machine Learning – Pre-College Research Program Aug. 2021, Jan. 2022
Gwangju Institute of Science and Technology
Summer & winter program instructor for "Physics Using Machine Learning" research program designed for advanced high school students through the Summer/Winter Pre-Undergraduate Research Participation Program (Pre-URP), under the supervision of Prof. Keun-Young Kim. Planned and designed the overall program, taught high school students the basics of machine learning, software skills (PyTorch), Neural ODE, and their application to classical mechanics problems.

As A Teaching Assistant

Introductory Physics I for Biological Sciences Laboratory Aug. - Dec. 2025
University of California, Merced
Led three laboratory sessions & graded lab notebooks. Supervisors: Prof. Antoinette Stone and Dr. Donna Jaramillo-Fellin.

Introductory Physics II for Biological Sciences Jun. - Aug. 2025
University of California, Merced
Led two discussion sessions and graded student assignments. Supervisor: Prof. John Wilson.

Cosmology Jan. - May 2024
University of California, Merced
Led the discussion session and graded student assignments. Supervisor: Prof. Anna Nierenberg.

Electrodynamics II Jan. - May 2024
University of California, Merced
Led the discussion session and graded student assignments. Occasionally, prepared and delivered lectures. Supervisor: Prof. Jing Xu.

Introductory Physics I for Biological Sciences JAN. - MAY 2021
University of California, Merced
Led two discussion sessions and graded student assignments. Supervisor: Prof. Antoinette Stone.

Introductory Physics I for Biological Sciences Laboratory AUG. - DEC. 2020
University of California, Merced
Led three laboratory sessions of bio-major students online using Beyond Labz, a simulation-based virtual laboratory due to COVID-19 restrictions. Supervisors: Prof. Carrie Menke and Dr. Kristina Callaghan.

History of the Universe and Humanity MAR. 2020 - JUN. 2020
Gwangju Institute of Science and Technology
Facilitated course progress and student discussions online. Supervisor: Prof. Keun-Young Kim.

Single Variable Calculus and Applications SEP. 2016 - DEC. 2016
Gwangju Institute of Science and Technology
Teaching assistant for undergraduate calculus I course under the supervision of Prof. Chi-Ok Hwang. Led Q&A sessions during office hours and graded students' homework.

PROFESSIONAL MEMBERSHIPS

American Physical Society 2020-2022, 2025-PRESENT

American Astronomical Society 2025-PRESENT

CERTIFICATES / ADDITIONAL TRAININGS

Cosmic Cartography with Roman:
Advances in Galaxy Structures, Distributions, Dark Matter, and Dark Energy JUL. 2025
Space Telescope Science Institute

JWST Summer School: High Redshift Transients with JWST AUG. 2025
Space Telescope Science institute

Advanced Pedagogy JAN. - MAY 2024
University of California, Merced - Teaching Commons

N3AS Summer School: Multi-Messenger Astrophysics JUL. 2023
University of California, Santa Cruz

UNIVERSITY SERVICE

Club Advisor 2024
University of California, Merced
Founding advisor for a new cultural student organization. Guided undergraduates in identifying the need for a Korean student club, drafting its mission, and establishing its structure. Mentored students in organizing community-building activities (e.g., Korean movie nights) that promoted cultural exchange on campus.

Campus Event Photographer 2025
University of California, Merced
Volunteered as an event photographer for various campus and departmental events, including the Diversity Forum and a public astronomy lecture & stargazing event. Photographs were featured on official university social media to document and promote the events.

OUTREACH EXPERIENCE

Skype A Scientist 2022

Participated in Skype A Scientist program, an outreach program toward the general public including students and families. Guided catapult design project at Crossing Park School (Alberta, Canada) virtually.

Science Communicator 2018 - 2020

Actively engaged in science communication initiatives, which included the design of science experiment shows, performing in these shows, and delivering science lectures related to my field of study and wider math and science topics.

SKILLS

Research	Mathematical Modeling and Simulation, Bayesian Inference, Uncertainty Quantification, Machine Learning, Data Analysis, Data Visualization, High Performance Computing	
Programming	Python (PyTorch, NumPy, SciPy)	(Advanced)
	MATLAB	(Advanced)
Data & Computing	Git, SQL, Cluster Computing (Slurm)	
Design Software	SolidWorks, Blender, Adobe Photoshop & Illustrator	
Teaching Tools	Canvas, Blackboard, Beyond Labz Virtual Lab	
Other Software	LaTeX, Microsoft Office	
Languages	English	(Fluent)
	Korean	(Native)

HONORS

- National scholarship to pursue B.S. in GIST, Korea Student Aid Foundation, 2015–2019
- Selected as one of the Top 10 Finalists in FameLab Korea, a three-minute science talk competition, 2018.
- Appointed as a Science Communicator by the Ministry of Science and ICT (MSIT) and the Korea Foundation for the Advancement of Science and Creativity (KOFAC), South Korea, 2018.
- Awarded Academic Excellence Scholarship, GIST, 2017–2018
- Awarded GIST Study Abroad Scholarship to Study at Caltech, 2019
- Awarded GIST Study Abroad Scholarship to Study at UC Berkeley, 2016

MEDIA COVERAGE

In English

[16] Emerging Technology from the arXiv. How to turn the complex mathematics of vector calculus into simple pictures. *MIT Technology Review*, Nov 2019. `technologyreview.com/s/614704/`.

In Korean

[17] 한소범 (So-Beom Han). ‘수포자’ 교수와 문학은 1도 모르던 물리학도, 천재시인 이상의 비밀을 풀다. *한국일보(Hankook Ilbo)*. `hankookilbo.com/News/Read/A2021100515390000378`.

[18] 선한결(Han-Gyeol Seon). 이상 ’건축무한육면각체’ 물리학으로 풀어... "이과가 이과했다". *한국경제(The Korea Economic Daily)*. `hankyung.com/it/article/202109230135i`.